Architecture: Diagram Legend

This legend defines **exclusive responsibility zones**.
Each concept appears **once** and only once.

🔴 Client / Entry Point

Shape: Rectangle **Color:** Red
Initiates requests into the system.
Examples

- Web / Mobile apps
- CLI tools
- External API consumers
- Webhooks

Rule: Clients never enforce platform policy or business rules.

🟠 Application / Orchestration Layer

Shape: Rounded Rectangle **Color:** Orange
Coordinates request handling and use-case execution. Knows *how* things happen, not *why*.
Examples

- Controllers
- BFF layers
- GraphQL resolvers
- API composition layers
- Workflow orchestrators

Explicitly NOT

- API gateways
- Security enforcement
- Infrastructure routing

🟡 Domain / Business Logic

Shape: Trapezoid Bottom (`trap-b`) **Color:** Yellow
Owns **business meaning and rules**.
Examples

- Domain services
- Bounded contexts (Tasks, Billing, Notifications-domain)
- Core business microservices
- Policy engines **(business policies only)**

Rule: If business rules change, this layer must change.

🟢 State / Persistence

Color: Green
Holds system state. Shape encodes access pattern.

Vertical Cylinder — Durable State

- SQL / NoSQL databases
- Event stores
- Object storage
- Data warehouses (storage role)

Horizontal Cylinder — Flowing State

- Queues (SQS, RabbitMQ)
- Streams (Kafka, Redis Streams)
- Background job queues
- Caches

🔵 Infrastructure / Traffic Management

Shape: Stadium **Color:** Blue

Controls **traffic flow and platform concerns**, but does not execute business logic.

Examples

- API gateways (Kong, Traefik, NGINX Ingress)
- Reverse proxies / load balancers
- Service mesh (Istio, Linkerd)
- CDN / edge routing
- CI/CD agents & runners
- Message brokers **as platform services**

Rule: Infra may route, throttle, or block — but never decide business outcomes.

🔷 Security / Platform Policy Enforcement

Shape: Hexagon **Color:** Dark Blue

Enforces **system-level safety and policy**.

Examples

- Identity providers (Auth0, Keycloak, Cognito)
- OAuth / JWT token services
- WAF
- Rate limiters
- Secrets management
- Circuit breakers & retries
- Platform policy enforcement points

Clarification: These are **platform policies**, not business policies.

🟣 Observability / Analytics / Cross-cutting

Shape: Reversed Parallelogram (`lean-r`) **Color:** Purple

Observes, measures, or analyzes the system **without influencing behavior**.

Examples

- Logging stacks (Loki, ELK)
- Metrics & tracing (Prometheus, Tempo, Jaeger)
- Monitoring dashboards
- Audit logs

- Telemetry pipelines
- BI / analytics platforms

⚪ External Services / Third-Party Systems

Shape: Rectangle (dashed border) **Color**: Grey
Represents systems outside the platform boundary that the application integrates with.
Examples

- Payment providers
- eSignature platforms
- Banking APIs
- Insurance providers
- Government registries
- Email / SMS delivery services

Rule: External systems are dependencies of the platform and are never controlled or governed by internal platform policies.

👤 Actor / Human Role

Shape: Rectangle (dashed border) **Color:** Light Red
Represents **a human role interacting with the system through a client application**.
Examples

- Anonymous Visitor
- Registered Customer
- Sales Agent
- Dealer Administrator
- Platform Administrator
- Support Engineer

Rule: Actors represent **people or organizational roles**, not software systems. They initiate interactions with the platform via **client interfaces** (web apps, mobile apps, APIs).

🎨 Mermaid Template (Authoritative)

```
%% Shapes
DOMAIN@{ shape: trap-b }
INFRA@{ shape: stadium }
SEC@{ shape: hex }
APP@{ shape: rounded}
DOMAIN@{ shape: trap-b }
DB@{ shape: cyl }
CACHE@{ shape: h-cyl }
OBS@{ shape: lean-r }

%% Classes
class CLIENT client
class INFRA infra
```

```
class SEC sec
class APP app
class DOMAIN domain
class DB storage
class CACHE storage
class OBS obs
class ACTOR actor
class EXTERNAL_SYSTEM external

%% Palette
classDef client fill:#ffe6e6,stroke:#d0021b,color:#424242;
classDef infra fill:#e6f0ff,stroke:#4a90e2,color:#424242;
classDef sec fill:#ccd9ff,stroke:#003399,color:#424242;
classDef app fill:#ffeacc,stroke:#ff6d00,color:#424242;
classDef domain fill:#fff2cc,stroke:#f5a623,color:#424242;
classDef storage fill:#e6f7e6,stroke:#7ed321,color:#424242;
classDef obs fill:#f9e6ff,stroke:#9013fe,color:#424242;
classDef actor fill:#fff0f0,stroke:#d0021b,stroke-dasharray:4 4,color:#424242;
classDef external fill:#f5f5f5,stroke:#6b6b6b,stroke-dasharray: 5
5,color:#424242;
```